

Research Concept Report

Prospective Bayesian Value-of-Information Guided Evolutionary Search

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Abstract

This proposed research investigates a new way of combining Bayesian reasoning, exploration–exploitation, and evolutionary or genetic algorithms. The central idea is to treat mutation not only as a mechanism for producing potentially better offspring, but also as an experiment whose result can teach the evolutionary algorithm how it should search in future generations.

Rather than asking only which candidate solution should be evaluated next, the proposed method considers which parent should be mutated, which variable should be changed, which mutation operator should be used, in which direction the change should occur, and how large the change should be. These evolutionary actions are evaluated according to the expected value of the information that their resulting offspring could provide for future population-level decisions.

The proposed contribution is therefore not simply a combination of Bayesian optimization and a genetic algorithm. Bayesian–evolutionary hybrids, surrogate-assisted evolutionary algorithms, guided mutation, and adaptive operator-selection techniques already exist. Instead, the candidate research contribution is a prospective decision-theoretic evolutionary framework in which variation actions are selected according to the expected Bayesian value of the information they produce for improving subsequent evolutionary search.

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1 Introduction

Evolutionary algorithms are powerful optimization methods for problems where gradients are unavailable, objective functions are nonlinear or multimodal, and the search space may contain many local optima.

A standard evolutionary algorithm maintains a population

$$P_t = \{\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_N\}, \quad (1)$$

where each individual represents a possible solution to an optimization problem of the form

$$\max_{\mathbf{x}} f(\mathbf{x}). \quad (2)$$

Through selection, mutation, crossover, and replacement, the population gradually evolves toward better solutions.

Mutation is particularly important because it provides exploration of the search space. However, conventional mutation is often stochastic. For example,

$$x'_i = x_i + \epsilon, \quad (3)$$

where

$$\epsilon \sim \mathcal{N}(0, \sigma^2). \quad (4)$$

The algorithm generally does not explicitly ask:

Why should this particular variable be mutated?

or

What information will be gained by evaluating this mutation?

The proposed research investigates whether mutation can instead be treated as a deliberate experiment.

The central question is:

Can an evolutionary algorithm choose mutations according to what it needs to learn, rather than only according to which offspring are predicted to have high immediate fitness?

2 Motivation

A conventional evolutionary algorithm can be summarized as

$$\text{Population} \rightarrow \text{Mutation} \rightarrow \text{Evaluation} \rightarrow \text{Selection}. \quad (5)$$

The proposed research introduces an additional reasoning stage:

$$\text{Population} \rightarrow \text{Bayesian belief} \rightarrow \text{Mutation experiment selection} \rightarrow \text{Evaluation} \rightarrow \text{Bayesian update} \rightarrow \text{Population} \quad (6)$$

The key idea is that an offspring can possess two different types of value:

$$\boxed{\text{Offspring Value} = \text{Immediate Fitness Value} + \text{Future Information Value}} \quad (7)$$

An offspring may therefore be useful even if its immediate fitness is poor.

For example, an offspring may help determine:

- whether a distant high-quality optimum exists,
- which variables are important,
- whether the current population is trapped in a local optimum,
- whether large or small mutations are appropriate,
- which mutation operator should receive more resources,
- which evolutionary lineage should be explored further.

The objective is therefore not simply to create good children.

The algorithm should sometimes create an offspring because **observing that offspring is strategically valuable**.

3 Relevant Existing Research

It is important to distinguish the proposed work from existing research.

The general concept of using learned information to guide evolutionary variation is not new.

Guided mutation methods have used probability models of promising regions to influence offspring generation. Surrogate-assisted evolutionary algorithms use approximate models of expensive objective functions to reduce the number of costly evaluations.

Other approaches use surrogate-predicted optima to guide mutation directions. Bayesian–evolutionary hybrid algorithms have also been proposed in which Bayesian optimization provides information that is transferred into an evolutionary algorithm.

Adaptive operator-selection methods dynamically determine which evolutionary operators should be used based on past performance or the current state of the search.

Therefore, the proposed novelty cannot simply be stated as

“A Bayesian model guides mutation.”

Nor is it sufficient to claim

“Bayesian uncertainty determines mutation magnitude.”

The proposed contribution must be more specific.

4 Bayesian Optimization

Bayesian optimization maintains a probabilistic belief about an unknown objective function.

After observing a dataset

$$D_t = \{(\mathbf{x}_1, y_1), \dots, (\mathbf{x}_t, y_t)\}, \quad (8)$$

a probabilistic model such as a Gaussian Process can provide

$$f(\mathbf{x}) \mid D_t \sim \mathcal{N}(\mu_t(\mathbf{x}), \sigma_t^2(\mathbf{x})). \quad (9)$$

Here,

$$\mu_t(\mathbf{x}) \quad (10)$$

represents expected performance, while

$$\sigma_t(\mathbf{x}) \quad (11)$$

represents predictive uncertainty.

This creates an exploration–exploitation problem.

Exploitation focuses on locations predicted to perform well.

Exploration focuses on regions where uncertainty is high or where an observation could be informative.

A Bayesian optimizer generally selects the next candidate using an acquisition function

$$\mathbf{x}_{t+1} = \arg \max_{\mathbf{x}} A(\mathbf{x}). \quad (12)$$

Possible acquisition functions include Expected Improvement, Upper Confidence Bound, Entropy Search, Predictive Entropy Search, and Knowledge Gradient methods.

5 Uncertainty Versus Value of Information

A fundamental distinction in this research is

$$\boxed{\text{Uncertainty} \neq \text{Value of Information}} \quad (13)$$

Uncertainty asks:

How little do we know about this location?

Information gain asks:

How much would an observation reduce uncertainty?

Value of Information asks:

How much would learning the result improve our future decisions?

A highly uncertain region may have little optimization value if no realistic observation from that region would affect future decisions.

Conversely, a moderately uncertain region may have very high value if resolving the uncertainty determines which search strategy should be followed.

This distinction is central to the proposed algorithm.

6 Proposed Core Idea

The proposed method transfers Value-of-Information reasoning from candidate evaluation to the **evolutionary variation mechanism itself**.

Instead of representing a decision only as a candidate solution

$$\mathbf{x}, \quad (14)$$

the method defines an evolutionary action

$$\boxed{a = (p, g, o, d, s)} \quad (15)$$

where

$$p = \text{parent selected for mutation}, \quad (16)$$

$$g = \text{gene or variable selected for mutation}, \quad (17)$$

$$o = \text{mutation operator}, \quad (18)$$

$$d = \text{mutation direction}, \quad (19)$$

$$s = \text{mutation magnitude}. \quad (20)$$

For example,

$$a_1 = (P_1, x_3, \text{Gaussian}, +, 0.2), \quad (21)$$

while

$$a_2 = (P_7, x_1, \text{Cauchy}, -, 5.0). \quad (22)$$

The algorithm evaluates not only the predicted quality of the resulting offspring, but also the expected information obtained from performing the evolutionary action.

The core question becomes:

If I perform this mutation and observe its result, how much better will I be able to manage future evolutionary search?

7 Illustrative Example

Consider a one-dimensional maximization problem.

The current evolutionary population is concentrated around

$$x = 40. \quad (23)$$

The best observed solution has fitness

$$f(40) = 80. \quad (24)$$

Based on the observations collected so far, the Bayesian model considers two hypotheses plausible.

7.1 Hypothesis A

There is one major optimum near

$$x = 45. \quad (25)$$

The posterior probability is

$$P(H_A) = 0.65. \quad (26)$$

7.2 Hypothesis B

The region around 40 is only a local optimum and a substantially better global optimum exists around

$$x = 70. \quad (27)$$

Its probability is

$$P(H_B) = 0.35. \quad (28)$$

The evolutionary algorithm considers three possible mutations:

$$M_1 : 40 \rightarrow 42, \quad (29)$$

$$M_2 : 40 \rightarrow 55, \quad (30)$$

$$M_3 : 40 \rightarrow 68. \quad (31)$$

Suppose the Bayesian model predicts the following fitness values.

<u>Table 1: Predicted outcomes of candidate mutations.</u>			
Mutation	H_A true	H_B true	Expected fitness
$M_1 : 40 \rightarrow 42$	84	82	83.3
$M_2 : 40 \rightarrow 55$	72	78	74.1
$M_3 : 40 \rightarrow 68$	30	110	58.0

From the perspective of immediate expected fitness,

$$M_1 > M_2 > M_3. \quad (32)$$

Therefore, a strongly fitness-oriented search method would naturally favor

$$M_1. \quad (33)$$

However, the important observation is that under the two hypotheses,

$$M_1 \quad (34)$$

has almost the same predicted result:

$$H_A \Rightarrow f(42) \approx 84, \quad (35)$$

while

$$H_B \Rightarrow f(42) \approx 82. \quad (36)$$

Therefore, evaluating M_1 provides little evidence about which hypothesis is correct.

Now consider M_3 .

If H_A is correct,

$$f(68) \approx 30. \quad (37)$$

If H_B is correct,

$$f(68) \approx 110. \quad (38)$$

Therefore, evaluating M_3 is highly diagnostic.

The mutation has poor expected immediate fitness but potentially very high information value.

This creates the central situation of interest:

$$\boxed{\text{Best predicted child} \neq \text{Best evolutionary experiment}} \quad (39)$$

8 A Simple Value-of-Information Calculation

Assume that after the current generation the algorithm must decide between two population strategies.

Let

$$L \tag{40}$$

represent allocating future population resources around the local region near 45.

Let

$$R \tag{41}$$

represent allocating resources around the remote region near 70.

Suppose

$$V(L, H_A) = 90, \tag{42}$$

$$V(L, H_B) = 82, \tag{43}$$

while

$$V(R, H_A) = 35, \tag{44}$$

$$V(R, H_B) = 115. \tag{45}$$

Without additional information, the expected value of choosing L is

$$E[V(L)] = 0.65(90) + 0.35(82) \tag{46}$$

$$= 58.5 + 28.7 \tag{47}$$

$$= 87.2. \tag{48}$$

The expected value of choosing R is

$$E[V(R)] = 0.65(35) + 0.35(115) \quad (49)$$

$$= 22.75 + 40.25 \quad (50)$$

$$= 63. \quad (51)$$

Therefore, without further information the rational decision is to select L .

Now suppose mutation M_3 is evaluated and its result almost perfectly distinguishes between the two hypotheses.

If H_A is true, the optimizer selects L and obtains value 90.

If H_B is true, the optimizer selects R and obtains value 115.

The expected value after obtaining this information is therefore

$$E[V \mid M_3] = 0.65(90) + 0.35(115) \quad (52)$$

$$= 58.5 + 40.25 \quad (53)$$

$$= 98.75. \quad (54)$$

The value of information of M_3 is therefore

$$VOI(M_3) = 98.75 - 87.2 \quad (55)$$

$$= 11.55. \quad (56)$$

Thus, although M_3 has relatively poor expected immediate fitness, the information it produces has substantial strategic value.

9 How the Proposed Method Differs from Bayesian Optimization

Bayesian optimization generally searches over candidate points:

$$\boxed{\mathbf{x}^* = \arg \max_{\mathbf{x}} A(\mathbf{x})} \quad (57)$$

where $\mathbf{x}$ is a location in the optimization domain.

The proposed method instead searches over evolutionary actions:

$$\boxed{a^* = \arg \max_a A_E(a)} \quad (58)$$

where

$$a = (p, g, o, d, s). \quad (59)$$

The search variable is therefore not simply the solution that should be evaluated.

It contains decisions about:

- which parent should reproduce,
- which gene should change,
- which mutation operator should be applied,
- which direction should be explored,
- how large the mutation should be.

More importantly, the information obtained from the offspring is valued according to whether it improves future **population-level evolutionary decisions**.

10 How the Proposed Method Differs from Adaptive Operator Selection

Adaptive Operator Selection (AOS) dynamically chooses evolutionary operators according to information obtained during the optimization process.

A simplified AOS mechanism can be represented as

$$\text{Use operator} \rightarrow \text{Observe offspring} \rightarrow \text{Assign reward} \rightarrow \text{Update operator probability}. \quad (60)$$

For example, if operator O_1 repeatedly produces successful offspring,

$$P(O_1) \quad (61)$$

may increase.

This can be interpreted as asking:

Which operator has worked well?

The proposed research asks a different prospective question:

Which operator should I deliberately use because observing its result would resolve an important uncertainty about how future evolution should proceed?

Therefore, a working distinction is

$$\boxed{\text{AOS: What has worked?}} \tag{62}$$

versus

$$\boxed{\text{Proposed method: What experiment is worth performing?}} \tag{63}$$

Modern AOS methods are sophisticated, so this distinction must be examined carefully in the literature review.

11 Two Forms of Learning

The proposed framework can potentially learn two different things.

11.1 Learning the Objective Landscape

The first uncertainty concerns

$$\boxed{\text{What does } f(\mathbf{x}) \text{ look like?}} \tag{64}$$

Examples include:

- whether another optimum exists,
- which variables strongly influence fitness,
- where promising regions are located,
- whether multiple separated basins exist.

11.2 Learning How Evolution Should Search

The second uncertainty concerns

$$\boxed{\text{How should the evolutionary process search this landscape?}} \tag{65}$$

Examples include:

- whether mutation should be aggressive or conservative,
- which mutation operator should dominate,

- which genes deserve more exploration,
- which lineage deserves more population resources,
- whether diversity should increase or decrease.

This second form of learning is especially important to the proposed research.

The algorithm is therefore potentially learning both

$$\boxed{\text{the optimization problem}} \tag{66}$$

and

$$\boxed{\text{how to evolve efficiently on that problem}}. \tag{67}$$

12 Proposed Mathematical Formulation

Let the population at generation t be

$$P_t. \tag{68}$$

Let the dataset of all observations collected so far be

$$D_t. \tag{69}$$

A Bayesian model maintains the posterior belief

$$B_t = p(f \mid D_t). \tag{70}$$

Consider a candidate evolutionary action

$$a = (p, g, o, d, s). \tag{71}$$

The action generates offspring

$$\mathbf{x}' = a(P_t). \tag{72}$$

Before evaluating the offspring, its possible observation is represented by

$$y \sim p(y \mid \mathbf{x}', D_t). \tag{73}$$

After observing y , the posterior becomes

$$B_{t+1}^{(a,y)} = p(f \mid D_t, (\mathbf{x}', y)). \quad (74)$$

Let

$$V(P, B) \quad (75)$$

represent the expected future optimization value that can be obtained from population P under Bayesian belief B .

The proposed evolutionary value of information is then conceptually

$$\boxed{VOI_E(a) = \mathbb{E}_y \left[V_{\text{future}} \left(P_t, B_{t+1}^{(a,y)} \right) \right] - V_{\text{future}}(P_t, B_t)} \quad (76)$$

This quantity asks:

How much better do we expect future evolutionary decisions to become after performing action a and observing the resulting child?

A complete action score could be

$$\boxed{A(a) = EI(a) + \beta VOI_E(a) - \lambda C(a)} \quad (77)$$

where

$$EI(a) = \text{expected immediate improvement}, \quad (78)$$

$$VOI_E(a) = \text{future evolutionary information value}, \quad (79)$$

$$C(a) = \text{cost or risk associated with the action}. \quad (80)$$

The selected action becomes

$$\boxed{a^* = \arg \max_a A(a)} \quad (81)$$

13 Exploration–Exploitation Interpretation

The formulation naturally creates an exploration–exploitation mechanism.

When the algorithm is uncertain and a mutation can resolve an important ambiguity,

$$VOI_E(a) \tag{82}$$

becomes large.

The evolutionary algorithm therefore performs exploratory mutations.

As the Bayesian model becomes more confident,

$$VOI_E(a) \rightarrow 0, \tag{83}$$

and immediate improvement becomes more important.

Thus,

$$A(a) \approx EI(a). \tag{84}$$

The algorithm becomes increasingly exploitative.

The resulting principle is

$$\boxed{\text{Explore while learning can still change important decisions.}} \tag{85}$$

This is different from using a predetermined schedule such as heavy exploration during the first 30 generations and exploitation afterward.

Exploration is driven by the remaining **decision-relevant uncertainty**.

14 Expected Distinguishing Behavior

A particularly important expected behavior is the deliberate creation of a predicted-inferior offspring.

Suppose a baseline algorithm generates offspring with

$$f(\mathbf{x}_{\text{baseline}}) = 85. \tag{86}$$

The proposed algorithm may intentionally generate

$$f(\mathbf{x}_{\text{proposed}}) = 55. \tag{87}$$

In the current generation, the proposed method appears worse.

However, the observation at the second point may resolve an uncertainty about the structure of the optimization landscape.

The algorithm may then:

- change its mutation operator,
- change mutation magnitude,
- shift population resources to another region,
- change the preferred evolutionary lineage,
- increase or reduce exploration.

After several generations,

$$f_{\text{best}}^{\text{proposed}}(t+k) > f_{\text{best}}^{\text{baseline}}(t+k). \quad (88)$$

Thus,

$$\boxed{\text{short-term sacrifice} \rightarrow \text{valuable information} \rightarrow \text{better long-term evolution}} \quad (89)$$

This behavior would provide one of the clearest demonstrations of the proposed mechanism.

15 Candidate Novelty

The novelty claim should remain narrow and carefully formulated.

Table 2: Distinction between established concepts and the proposed contribution.

Concept	Status
Bayesian optimization combined with evolutionary algorithms	Existing research
Probability models guiding mutation	Existing research
Surrogate models guiding mutation directions	Existing research
Bayesian information adapting mutation magnitude	Existing research
Adaptive selection among evolutionary operators	Existing research
Using uncertainty to select expensive evaluations	Existing research
Value-of-information reasoning in Bayesian optimization	Existing research
Prospective VOI over a complete evolutionary variation action	Candidate contribution
Valuing information according to its effect on future population-level evolutionary decisions	Candidate contribution
Deliberately generating predicted-inferior offspring because they improve future evolutionary policy	Candidate distinguishing behavior

A suitable working novelty statement is:

We propose a prospective Bayesian evolutionary framework in which variation is represented as an action over parent, variables, mutation operator, direction, and magnitude. Rather than selecting variation actions solely according to predicted offspring fitness, model uncertainty, or historical operator performance, the method evaluates the expected value of the observation generated by each action in terms of how that observation can improve subsequent population-level evolutionary decisions.

This should initially be considered a **novelty hypothesis**, rather than a confirmed claim of being the first method of its type.

16 Research Questions

The primary research question is:

Can evolutionary variation be formulated as Bayesian experimental design, such that an evolutionary algorithm chooses mutations according to the expected value of the information their offspring provide for future population-level search decisions?

Several supporting research questions follow.

- RQ1:** Can prospective Value-of-Information reasoning improve evolutionary optimization efficiency?
- RQ2:** Can the algorithm identify situations where a predicted-inferior offspring is strategically valuable?
- RQ3:** Can information obtained from one offspring improve operator selection for future generations?
- RQ4:** Can the algorithm automatically transition from exploration to exploitation as decision-relevant uncertainty decreases?
- RQ5:** Does prospective VOI outperform simple uncertainty-based mutation?
- RQ6:** Does prospective VOI outperform mutation strategies based only on historical operator performance?
- RQ7:** Under what classes of optimization problems is the proposed mechanism most beneficial?

17 Research Hypothesis

A testable research hypothesis is:

On optimization problems containing structural uncertainty, deceptive local optima, expensive evaluations, or competing plausible search hypotheses, prospective Value-of-Information-guided evolutionary variation will require fewer expensive evaluations and achieve better long-horizon optimization performance than evolutionary methods that select mutation based only on immediate fitness, predictive uncertainty, or historical operator performance.

18 Proposed Algorithm

A preliminary algorithm can be written as follows.

Step 1: Initialize Population

Generate

$$P_0 = \{\mathbf{x}_1, \dots, \mathbf{x}_N\}. \quad (90)$$

Evaluate all initial individuals.

Construct

$$D_0 = \{(\mathbf{x}_i, f(\mathbf{x}_i))\}_{i=1}^N. \quad (91)$$

Step 2: Fit Bayesian Model

Estimate

$$p(f \mid D_t). \quad (92)$$

A Gaussian Process would provide a convenient starting model.

Step 3: Generate Candidate Evolutionary Actions

Construct candidate actions

$$a_j = (p_j, g_j, o_j, d_j, s_j). \quad (93)$$

Step 4: Predict Possible Outcomes

For every candidate action, estimate

$$p(y \mid a_j, D_t). \quad (94)$$

Step 5: Estimate Immediate Improvement

Calculate

$$EI(a_j). \quad (95)$$

Step 6: Estimate Evolutionary Value of Information

Calculate

$$VOI_E(a_j). \quad (96)$$

Step 7: Calculate Action Utility

$$A(a_j) = EI(a_j) + \beta VOI_E(a_j) - \lambda C(a_j). \quad (97)$$

Step 8: Select Mutation Action

$$a^* = \arg \max_{a_j} A(a_j). \quad (98)$$

Step 9: Generate and Evaluate Offspring

Apply

$$a^* \quad (99)$$

to generate offspring

$$\mathbf{x}'. \quad (100)$$

Evaluate

$$y' = f(\mathbf{x}'). \quad (101)$$

Step 10: Update Bayesian Model

Add the new observation:

$$D_{t+1} = D_t \cup \{(\mathbf{x}', y')\}. \quad (102)$$

Update

$$p(f \mid D_{t+1}). \quad (103)$$

Step 11: Update Population and Evolutionary Policy

Use the new observation to update:

- population membership,
- mutation operator probabilities,
- mutation magnitudes,
- preferred genes or variables,
- evolutionary lineages,
- exploration intensity.

Repeat until the stopping criterion is reached.

19 Experimental Methodology

The first study should deliberately avoid robotics.

The purpose is to determine whether the proposed optimization mechanism has value independently of any particular application.

The algorithm should first be tested on controlled synthetic optimization problems.

Useful benchmark classes include:

- multimodal functions,
- deceptive functions,
- noisy objective functions,
- expensive black-box optimization,

- high-dimensional problems,
- functions with strong variable interactions,
- landscapes containing separated global and local optima.

Artificial benchmark problems should also be specifically constructed to create competing hypotheses similar to the H_A/H_B example.

These experiments can test whether the algorithm performs strategically valuable exploration.

20 Baseline Algorithms

The proposed method should be compared against:

1. Standard Genetic Algorithm,
2. Adaptive Genetic Algorithm,
3. Differential Evolution,
4. CMA-ES,
5. Bayesian Optimization,
6. Knowledge-Gradient-based Bayesian Optimization,
7. Surrogate-Assisted Evolutionary Algorithms,
8. Adaptive Operator Selection,
9. uncertainty-directed mutation,
10. the proposed prospective VOI method.

The objective is not merely to show that the proposed method obtains a better final solution.

The experiments should show **why** its decisions differ from those of existing algorithms.

21 Ablation Study

Ablation experiments are necessary to determine which component produces the performance improvement.

The full algorithm is

$$A(a) = EI(a) + \beta VOI_E(a). \quad (104)$$

It should be compared against the following variants.

21.1 Immediate Improvement Only

$$A(a) = EI(a). \quad (105)$$

21.2 Predictive Uncertainty Only

$$A(a) = \sigma(a). \quad (106)$$

21.3 Information Gain Only

$$A(a) = IG(a). \quad (107)$$

21.4 Historical Operator Reward

$$A(a) = Q(o), \quad (108)$$

where $Q(o)$ represents the historical success of operator o .

21.5 Random Mutation

Mutation actions are chosen according to the conventional mutation distribution.

The key experimental question is whether

$$VOI_E \quad (109)$$

provides information beyond ordinary uncertainty, information gain, and historical operator performance.

22 Evaluation Metrics

Possible evaluation metrics include:

- best fitness versus number of expensive evaluations,
- final solution quality,
- cumulative regret,
- simple regret,
- probability of discovering the global optimum,

- generations required to escape local optima,
- information gained per evaluation,
- number of unnecessary evaluations,
- population diversity,
- computational overhead,
- robustness across random seeds.

Because the proposed method is intended primarily for situations in which evaluations are expensive, performance should be analyzed primarily as a function of

$$\boxed{\text{number of expensive objective evaluations}} \quad (110)$$

rather than only number of generations.

23 Connection to Localization-Aware Robotics

The proposed optimization method has a natural connection to localization-aware goal-directed exploration.

In the current navigation research, environmental geometry can be related to localization performance.

A future robot could instead maintain a Bayesian model

$$p(R_{\text{loc}} \mid \mathbf{z}, D_t), \quad (111)$$

where R_{loc} represents localization risk and $\mathbf{z}$ could contain features such as

$$\mathbf{z} = [\lambda_{\min}(F), \det(F), \text{tr}(F), d_{\text{wall}}, v, \omega, q_{\text{scan}}, \dots]. \quad (112)$$

Possible terms include:

- Fisher Information metrics,
- wall distance,
- geometric structure,
- scan-matching quality,
- robot velocity,

- angular velocity,
- pose covariance,
- SLAM state.

The robot would not assume that the relationship between geometry and localization degradation is perfectly known.

Instead, it would learn this relationship online.

A candidate trajectory could be represented by

$$\pi = [w_1, w_2, \dots, w_H], \quad (113)$$

where the w_i are trajectory waypoints.

Evolutionary actions could modify:

- waypoint locations,
- route branches,
- robot velocity,
- exploration detours,
- trajectory shape.

Suppose a wall-following route is known to be safe while a shorter route through an open region is uncertain.

A conventional localization-aware planner may avoid the open region.

The proposed method could instead create a controlled mutation that causes the robot to partially probe the uncertain area if the observation is expected to have sufficiently high future value.

If localization remains reliable, the Bayesian posterior would shift toward lower predicted localization risk.

If localization becomes poor, the posterior would shift toward higher predicted risk.

Future evolutionary planning would then change accordingly.

The robot would therefore learn not only

$$\text{where to navigate,} \quad (114)$$

but

when it is worth performing a controlled experiment in order to improve future navigation decisions.

(115)

24 Potential Long-Term Research Direction

The broader research vision is to develop an autonomous evolutionary system that learns both

the problem

(116)

and

how to search the problem

.
(117)

Mutation would no longer be interpreted only as random noise.

Instead,

Mutation = Experiment

(118)

Each offspring represents a hypothesis about a potentially useful region or evolutionary strategy.

Each evaluation produces an observation.

Bayesian inference updates the system's beliefs.

Evolution uses the new knowledge to determine which experiments should be performed next.

The overall loop becomes

Hypothesize → Experiment → Observe → Learn → Evolve

(119)

and then repeats.

25 Expected Scientific Contribution

If successful, the proposed research could contribute at several levels.

25.1 Algorithmic Contribution

A new evolutionary variation framework based on prospective Bayesian Value of Information.

25.2 Decision-Theoretic Contribution

A formulation in which the information value of an evolutionary action is defined according to its effect on future population decisions.

25.3 Exploration–Exploitation Contribution

An exploration mechanism driven by decision-relevant uncertainty rather than a manually defined exploration schedule.

25.4 Evolutionary Learning Contribution

A framework in which evolution learns not only candidate solutions, but also how mutation and search strategies should adapt to the optimization landscape.

25.5 Robotics Contribution

A future application in which localization-aware autonomous navigation can safely investigate uncertain environmental regions when the expected long-term information value justifies the exploratory action.

26 Risks and Open Questions

Several important challenges must be addressed.

26.1 Computational Cost

Estimating

$$VOI_E(a) \tag{120}$$

for many candidate evolutionary actions may itself be expensive.

Approximate methods may therefore be necessary.

26.2 Defining Future Population Value

The quantity

$$V_{\text{future}}(P, B) \tag{121}$$

must be defined carefully.

It may be impossible to compute exact multi-generation value, requiring short-horizon approximations.

26.3 Scalability

The number of possible actions can grow rapidly because the action space contains parent, gene, operator, direction, and magnitude.

Efficient action-space reduction will therefore be important.

26.4 Model Error

A Gaussian Process or other Bayesian surrogate may itself be inaccurate, especially in high-dimensional spaces.

26.5 Novelty Boundary

Neighboring fields include:

- Bayesian optimization,
- Knowledge Gradient,
- Bayesian experimental design,
- surrogate-assisted evolutionary optimization,
- adaptive operator selection,
- estimation-of-distribution algorithms,
- hyperheuristics,
- reinforcement-learning-assisted evolutionary algorithms,
- active learning.

A systematic literature review is therefore essential before making a strong claim of first novelty.

27 Recommended Research Sequence

The research can be divided into several stages.

Stage 1: Literature Review

Identify the closest methods in Bayesian optimization, adaptive operator selection, surrogate-assisted evolutionary computation, and Bayesian experimental design.

Stage 2: Formal Toy Problem

Develop a simple one-dimensional or two-dimensional example where immediate fitness and future information value produce clearly different mutation decisions.

Stage 3: Mathematical Formulation

Formally define

$$VOI_E(a). \tag{122}$$

Stage 4: Prototype Algorithm

Implement the method using a simple evolutionary algorithm and Gaussian Process model.

Stage 5: Synthetic Benchmarks

Evaluate the method on deceptive and multimodal benchmark functions.

Stage 6: Comparison and Ablation

Compare against conventional evolutionary algorithms, Bayesian optimization, surrogate-assisted evolution, and adaptive operator selection.

Stage 7: Robotics Application

Apply the algorithm to localization-aware goal-directed exploration.

28 Conclusion

The proposed research investigates whether evolutionary mutation can be reformulated as a Bayesian experimental-design problem.

A conventional evolutionary algorithm effectively asks:

Generate another candidate and determine whether it is better.

A model-guided evolutionary algorithm asks:

Generate a candidate that the current model predicts may be better.

Bayesian optimization asks:

Which candidate should be evaluated next given both predicted performance and uncertainty?

The proposed evolutionary framework instead asks:

Which evolutionary action should be executed because the observation produced by its offspring will be most valuable for deciding how future generations should evolve?

The most distinctive expected behavior is that the method may intentionally create an offspring with poor predicted immediate fitness because learning from that offspring is expected to produce better descendants in later generations.

The proposed concept can therefore be summarized as

Mutation as Bayesian Experimental Design

(123)

with the broader research question:

Can an evolutionary algorithm intentionally perform a short-term inferior experiment because the resulting information allows the population to make superior long-term evolutionary decisions?

This represents the central concept to be mathematically formalized, systematically compared with existing work, and evaluated as a potential novel contribution.

References

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- [4] Literature on the Knowledge Gradient for Bayesian optimization and decision making under uncertainty. Knowledge Gradient methods quantify the expected value that an additional measurement provides for subsequent decision making.
- [5] Literature on Adaptive Operator Selection for evolutionary algorithms and metaheuristics. These methods dynamically adjust operator selection based on operator performance, optimization state, or learned policies.